

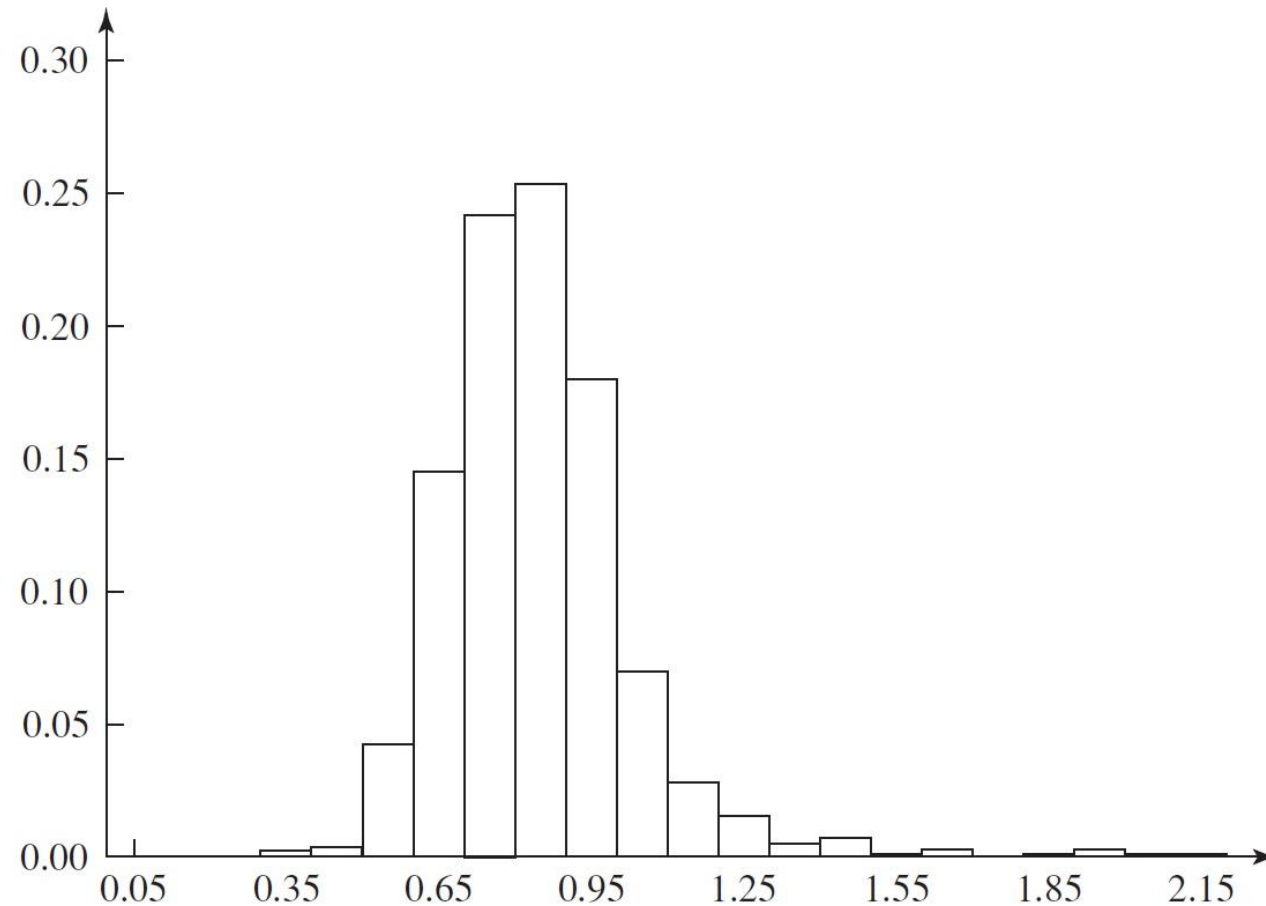
Simulation Modeling and Analysis

Dr. Md. Nasim Adnan

Introduction

- To carry out a simulation using random inputs such as interarrival times or demand sizes, we have to specify their probability distributions.
- For example, in the simulation of the single-server queueing system, the interarrival times can be taken to be IID exponential random variables with a mean of 1 minute; the demand sizes in the inventory simulation were specified to be 1, 2, 3, or 4 items with respective probabilities $1/6$, $1/3$, $1/3$, and $1/6$.
- Given that the input random variables to a simulation model follow particular distributions, **the simulation proceeds through time** by generating random values from those particular distributions.

A histogram for 856 ship-loading times (in days)



USEFUL PROBABILITY DISTRIBUTIONS

- The purpose of this study is to discuss a variety of distributions that have been found to be useful in simulation modeling.
 - Continuous distributions
 - Discrete distributions

Continuous Distributions: Uniform

Continuous Distributions: Exponential

Discrete Distributions: Bernoulli